1 Communication System

Objective

After completing this chapter, user will be able to understand:

- Basics of signals
- Basic operations on signals
- Frequency domain representation of signals
- Few important frequency transforms

Introduction

- Just as a carpenter requires a proper set of tools before he can sit down to make a piece of furniture, in a similar manner, a communication engineer needs to know about signals before he can start the process of learning communication.
- Signal processing operations were performed.
- The device that performs these operations.
- The underlying physics
- Thus, to study the basic form of modulation and signal processing used in communication, it will be fruitful to have a quick review of the concepts of signals and systems.

Basics of Signals

Definitions

A signal is the mathematical representation of a physical phenomenon whicth contains some information.

Physical phenomenon: AC voltage

Mathematical representation: $x(t) = A\sin\omega_0 t$

$f(x) \rightarrow \begin{cases} \rightarrow \textit{Dependent variable:f}(x) \\ \rightarrow \textit{Independent variable: x} \end{cases}$

Important categories of signals:

1. $X(t)$
   - $\rightarrow$ $DV : X(t)$
   - $\rightarrow$ $IDV : t \rightarrow Time$
   - $\rightarrow$ *Time* domain signals

2. $X(n)$
   - $\rightarrow$ $DV : X(n)$
   - $\rightarrow$ $IDV : n \rightarrow Time$ (Discrete Time) $\rightarrow$ *Integer*
   - $\rightarrow$ "*Discrete* Time Domain Signals"

3. $X(\omega)$ or $X(f)$
   - $\rightarrow$ $DV : X(\omega)$ or $X(f)$
   - $\rightarrow$ $IDV : \omega$ or f
   - $\rightarrow$ *Frequency* domain signals

$\omega \rightarrow$ Analog frequency (rad/sec)

$f \rightarrow$ Linear frequency (Hz) or $\frac{1}{\text{sec}}$

$$\underbrace{\omega}_{\frac{rad}{\sec}} = 2\pi \underbrace{f}_{\frac{1}{\sec}}$$

Example: $\underbrace{\omega_i(t)}_{\frac{rad}{\sec}} = \underbrace{\omega}_{\frac{rad}{\sec}} + \underbrace{k_f}_{\frac{rad}{V-\sec}} \underbrace{m(t)}_{Volt}$

$$2\pi f_i(t) = 2\pi f_c + k_f m(t)$$

$$\underset{Hz}{f_i(t)} = \underset{Hz}{f_c} + \frac{k_f}{2\pi}\underbrace{m(t)}_{V} \qquad k_f = \frac{rad}{V-\sec}$$

$$\frac{k_f}{2\pi} = \frac{rad}{rad\,(v-\sec)}$$

$$\frac{k}{2\pi} = \frac{1}{V-\sec}$$

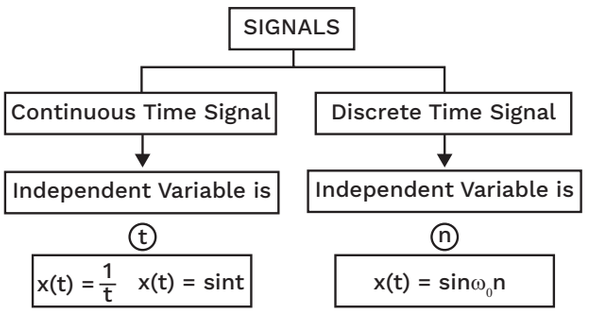
Example: $\underbrace{f_i}_{Hz} = \underbrace{f_c}_{Hz} + \underbrace{k_f}_{\frac{Hz}{Volt}} \underbrace{m(t)}_{Volt}$

$$2\pi f_i(t) = 2\pi f_c + 2\pi k_f m(t)$$

$$\underset{\frac{rad}{\sec}}{\omega_i(t)} = \underset{\frac{rad}{\sec}}{\omega_c} + \underset{rad}{2\pi}\ \underset{\frac{Hz}{Volt}}{k_f}\ \underbrace{m(t)}_{Volt} \rightarrow \text{rad/sec}$$

4. $X(\lambda)$
   - $\rightarrow$ $DV : X(\lambda)$
   - $\rightarrow$ $IDV : \lambda \rightarrow$ Sample point
   - $\rightarrow$ *Random* process

5. $X(\lambda, t)$
   - $\rightarrow$ $DV : X(\lambda, t)$
   - $\rightarrow$ $IDV : \lambda, t$
   - $\rightarrow$ *Random* process

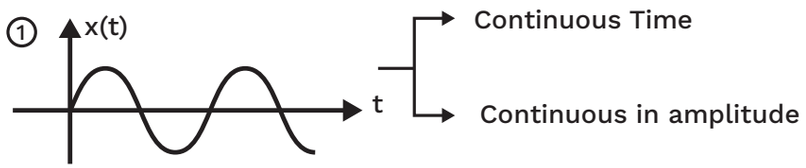


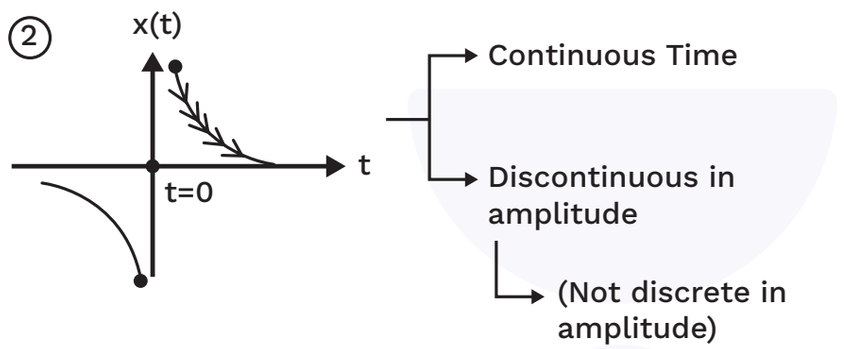
Note: A signal is continuous: *'Independent Variable = t'*

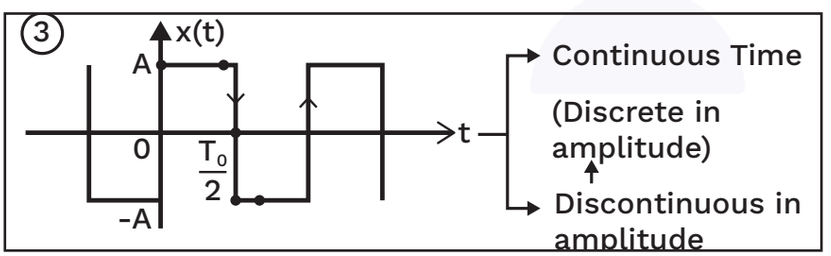
1. Continuous in time → x(t)
2. Continuous in amplitude →

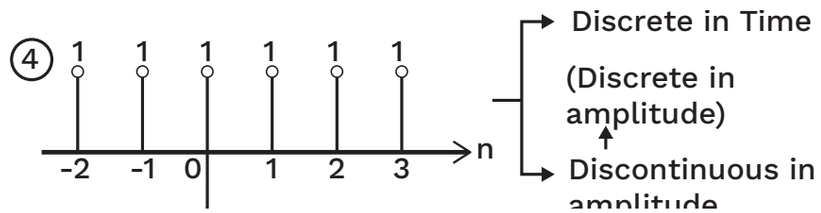
At every instant of time

LHL = RHL = Functional value = Finite









Analog and Digital Signals

Analog signals:

S.1 $-\infty < t < \infty$ $-\infty < n < \infty$

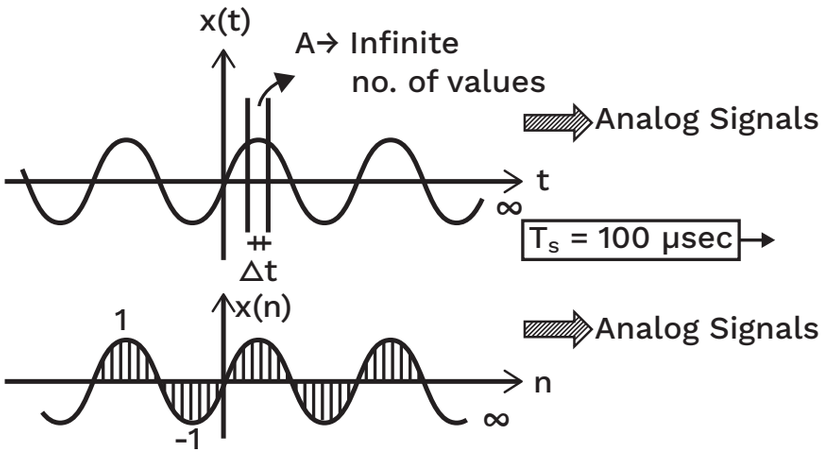
S.2 Amplitude containing infinite no. of values → Analog signals

Digital signals:

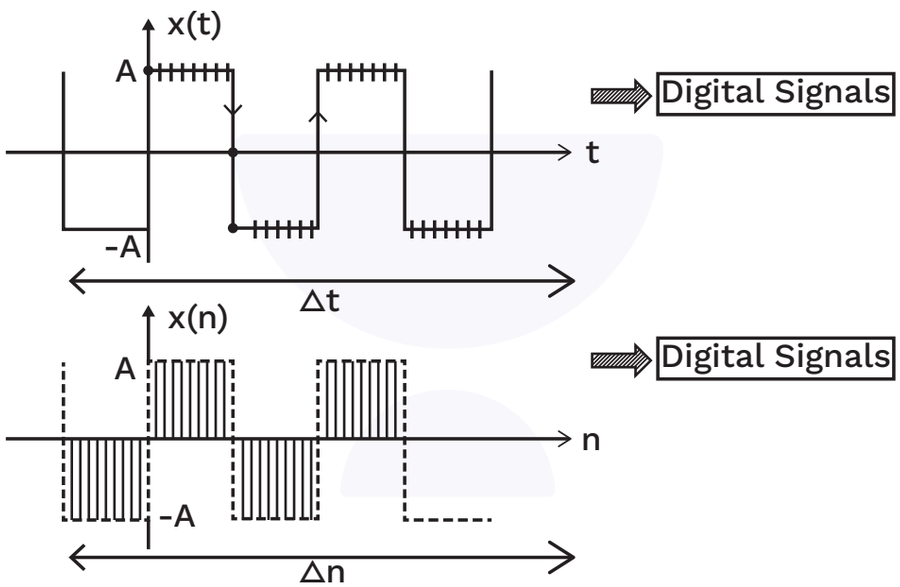
S.1 $-\infty < t < \infty$ $-\infty < n < \infty$

S.2 Amplitude containing finite no. of values → Digital signals

Example:



Example:

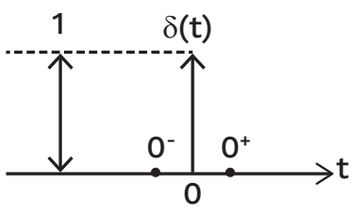


Note:

1. Sampling converts CTS into DTS.
2. *'Sampling does not affect analog or digital nature of signal provided* f_s *and* T_s *is optimum'.*

Standard Continuous Time Signals

Unit impulse signal:



$$\begin{cases} \delta(t) = 0 : t \neq 0 \\ \delta(t) \neq 0 : t = 0 \rightarrow \delta(t) \rightarrow \infty \end{cases}$$

$$\int_{\infty^-}^{\infty} \delta(t)dt = \int_{0^-}^{0^+} \delta(t)dt = 1$$

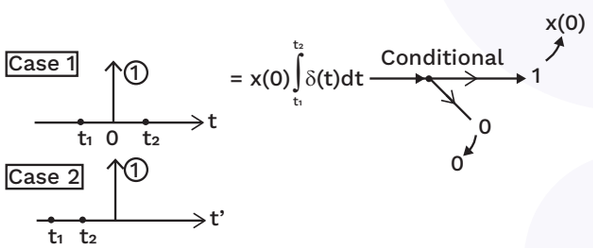
Properties:

1. $\delta(-t)=\delta(t)$
2. $\delta(at)=\frac{1}{|a|}\delta(t)$
3. $\delta(-at+b)=\delta\left(-a\left(t-\frac{b}{a}\right)\right)=\frac{1}{|-a|}\delta\left(t-\frac{b}{a}\right)$
4. $\delta(-at-b)=\delta\left(-a\left(t+\frac{b}{a}\right)\right)=\frac{1}{|-a|}\delta\left(t+\frac{b}{a}\right)$
5. $x(t)\delta(t)=x(0)\delta(t)$
6. $x(t)\delta(-at+b)=x(t)\delta\left(-a\left(t-\frac{b}{a}\right)\right)$

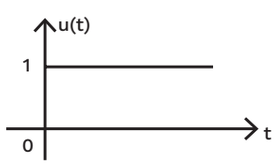
   $=x(t)\frac{1}{|-a|}\delta\left(t-\frac{b}{a}\right)=\frac{1}{|-a|}x\left(\frac{b}{a}\right)\delta\left(t-\frac{b}{a}\right)$

   $\downarrow$

   $t=\frac{b}{a}$
7. $\int_{t_1}^{t_2}x(t)\delta(t)dt=\int_{t_1}^{t_2}x(0)\delta(t)dt$



Unit step signal:



$$u(t)=\begin{cases}1 & t\geq 0\\ 0 & t<0\end{cases}$$

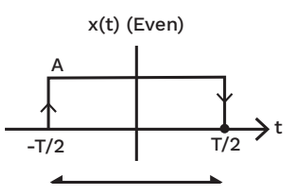
Unit ramp signal:

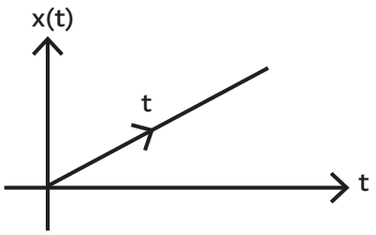
$$x(t)=tu(t)=r(t)=\begin{cases}t & t\leq 0\\ 0 & t<0\end{cases}$$

Some Advanced Signals

Rectangular function:

1. $x(t)=\begin{cases}A & |t|\leq\frac{T}{2}\quad\left(\frac{-T}{2}\leq t\leq\frac{T}{2}\right)\\ 0 & \text{elsewhere}\end{cases}$

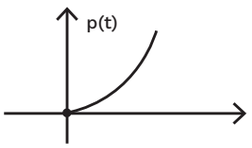




Unit parabolic signal:

$$p(t)=\frac{t^2}{2}u(t)$$

$$p(t-1)=\frac{(t-1)^2}{2}u(t-1)$$



Operations on CTS

Time shifting:

Given: $x(t)$ vs t

Plot: $x(t-t_0)$ vs t $\rightarrow$ Shift rightwards by t_0

Time scaling:

Given: $x(t)$ vs t

Plot: $x(at)$ vs t $\rightarrow$ Time axis is divided by a.

Time reversal:

Given: $x(t)$ vs t

Plot: $x(-t)$ vs t $\rightarrow$ *'Mirror image w.r.t. vertical axis'.*

Combined operations:

$x(-at+b)\Rightarrow Shift\rightarrow Scale\rightarrow$ Reversal

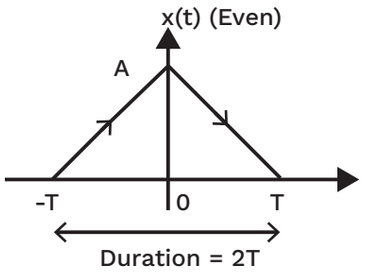
$$x(t)\xrightarrow[\text{Shifting}]{}x(t+b)\xrightarrow[\text{Scaling}]{}x(at+b)$$

$$\xrightarrow[\text{Reversal}]{}x(-at+b)$$

2. $x(t) = \overset{\text{Amplitude}}{\overset{\uparrow}{A}} \text{ rect } \left(\frac{t}{\underset{\text{Duration}}{\underset{\uparrow}{T}}}\right)$

3. $x(t) = Au\left(t + \frac{T}{2}\right) - Au\left(t - \frac{T}{2}\right)$

Triangular function:



1. $x(t) = \begin{cases} A\left(1 - \frac{|t|}{T}\right) & \overset{-T \le t \le T}{\overset{\uparrow}{|t| \le T}} \\ 0 & \text{elsewhere} \end{cases}$

2. $x(t) = \overset{\text{Peak value}}{\overset{\uparrow}{A}} \text{ tri } \left(\frac{t}{\underset{\frac{\text{Duration}}{2}}{\underset{\uparrow}{T}}}\right)$

Sinc functions:

$\text{sinc}(t) = \frac{\sin \pi t}{\pi t}$ $\qquad \pi \frac{\sin 6t}{3t} = 2\pi \text{ sinc}\left(\frac{6t}{\pi}\right)$

$\text{sinc}(kt) = \frac{\sin(k\pi t)}{k\pi t}$

$\frac{\sin at}{bt} \longrightarrow \frac{a}{b}\text{sinc}\left(\frac{at}{\pi}\right)$

Properties of sinc (t):

1. $\text{sinc}(-t) = \text{sinc}(t) \rightarrow$ Even

2. $\lim_{t \to 0} \frac{\sin t}{t} = 1$ $\qquad -1 \le \sin\theta \le 1$

3. $\lim_{t \to \pm\infty} \frac{\overset{-1 \le \le 1 \rightarrow \text{finite}}{\overbrace{\sin t}}}{\underset{\pm\infty}{\underbrace{t}}} = 0$

4. Zero cross-over time instances:

$\text{sinc}(t) = 0$ $\qquad$ If $n = 0 : \rightarrow t = 0$

$\frac{\sin \pi t}{\pi t} = 0 \rightarrow$ $\qquad \lim_{t \to 0} \text{sinc}(t) = 1$

$\sin \pi t = 0 = \sin n\pi$

$\pi t = n\pi \quad n \in I \Rightarrow n = 0,\ 1, 2, 3...$

$t = n \qquad\qquad -1, -2, -3...$

$\text{sinc}(t) \longrightarrow$ Zero cross-over time instant:

$t = n : \ n \in I$

$n \ne 0$

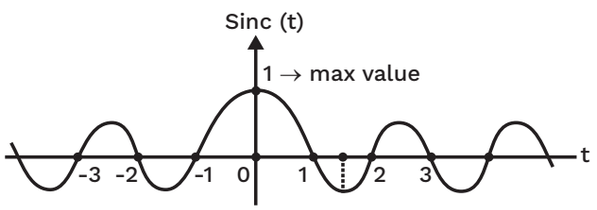
$\text{sinc}(f) \longrightarrow$ Zero cross-over frequencies

$f = n\text{Hz} \ \ n \in I$

$n \ne 0$

$\text{sinc}(kt) \longrightarrow kt = n \quad n \in I$

$n \ne 0$



HW.: $\left[\text{sinc}(t)_{\min}\right]$ at t = ?

Sampling function:

$x(t) = \frac{\sin t}{t} = 1 \cdot \text{sinc}\left(\frac{t}{\pi}\right) = Sa[t]$

1. Even

2. $\lim_{t \to 0} Sa(t) = 1$

3. $\lim_{t \to \pm\infty} Sa(t) = 1$

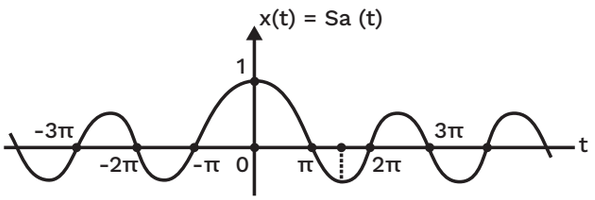
4. Zero cross-over

$\frac{t}{\pi} = n \quad n \in I \quad n \ne 0$

$t = n\pi \quad n \in I \quad n \ne 0$

$$\frac{\sin at}{bt} = \frac{a}{b} sa[at]$$

$$\pi \frac{\sin 6t}{4t} = \frac{3\pi}{2} sa[6t]$$



Example: Calculate zero cross-over frequencies in Hz.

$$X(\omega) = \frac{\sin 2\omega}{\omega}$$

$$X(\omega) = 2Sa[2\omega]$$

Zero cross-overs:

$$2\omega = n\pi \quad n \in I$$

$$n \neq 0$$

$$4\pi f = n\pi$$

$$f = \frac{n}{4} \text{ Hz} \qquad n \neq 0$$

$$n \in I$$

Energy and Power Signals

Energy signal: x(t)

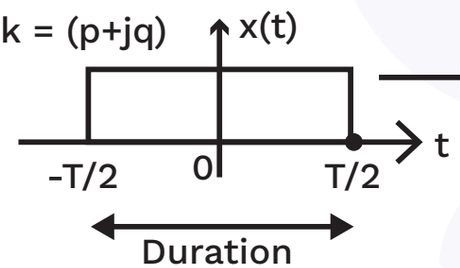
Step 1: $|x(t)|$

Step 2: $|x(t)|^2$

Step 3: $E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$

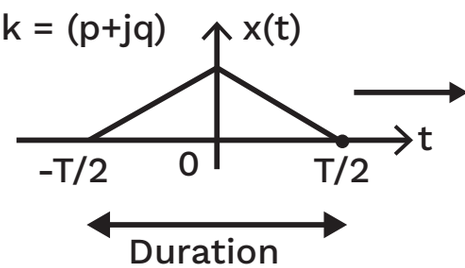
Energy of some standard signals:

① k = (p+jq)



$E_x = (p^2+q^2) \times T = |k|^2 \times T$

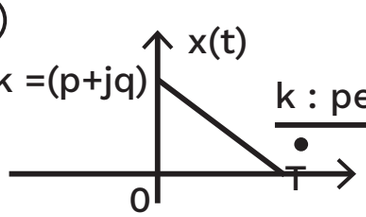
② k = (p+jq)



k = peak value

$$Ex = \frac{|k|^2}{3} \times T = \frac{(p^2+q^2)}{3} \times T$$

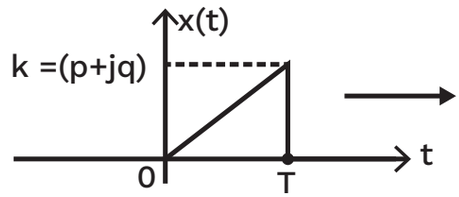
③



k : peak value

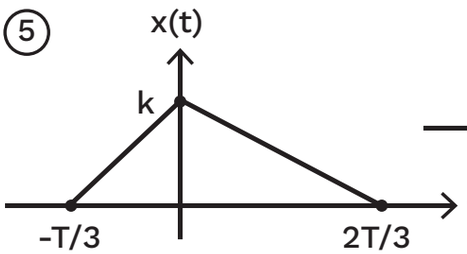
$$Ex = \frac{|k|^2}{3} \times T = \frac{(p^2+q^2)}{3} \times T$$

④



$$\rightarrow Ex = \frac{|k|^2}{3} \times T = \frac{(p^2+q^2)}{3} \times T$$

⑤



$$\rightarrow \boxed{Ex = \frac{|k|^2}{3} \times \text{duration}}$$

Let $x(t) = x_1(t) + x_2(t)$ $\quad |A+B|^2 = |A|^2 + |B|^2 + AB^* + A^*B$

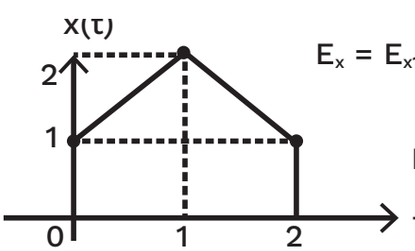
$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E_x = \int_{-\infty}^{\infty} |x_1(t) + x_2(t)|^2 dt = \int_{-\infty}^{\infty} |x_1(t)|^2 dt + \int_{-\infty}^{\infty} |x_2(t)|^2 dt$$
$$+ \int_{-\infty}^{\infty} x_1(t) x_2^*(t) dt + \int_{-\infty}^{\infty} x_1^*(t) x_2 x(t) dt$$

$$E_x = E_{x_1} + E_{x_2} + \underbrace{\int_{-\infty}^{\infty} x_1(t) x_2^*(t) dt}_{0} + \underbrace{\int_{-\infty}^{\infty} x_1^*(t) x_2(t) dt}_{0}$$

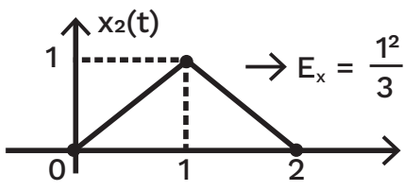
$x_1(t)$ and $x_2(t) \rightarrow$ Orthogonal

$$E = E_{x_1} + E_{x_2}$$

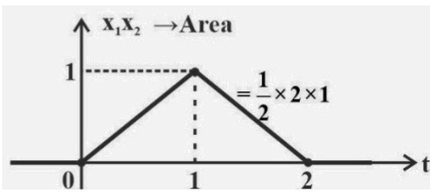


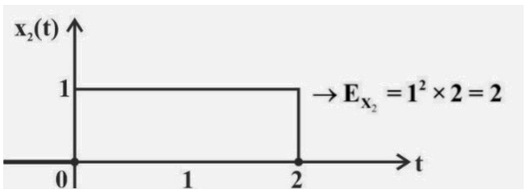
$$E_x = E_{x1} + E_{x2} + 2 \int_{-\infty}^{\infty} x_1(t) x_2(t) dt$$

$$E_x = \frac{2}{3} + 2 + 2 \int_{-\infty}^{\infty} x_1(t) x_2(t) dt$$



$$\rightarrow E_x = \frac{1^2}{3} \times \frac{2}{3}$$





$$E_x=\frac{2}{3}+2+2=\frac{14}{3}\text{ Joules}$$

Important points:

$$x(t)\longrightarrow E_x$$

$$x(t-a)\longrightarrow E_x$$

$$x(-t)\longrightarrow E_x$$

$$x(at)\longrightarrow\frac{E_x}{|a|}$$

$$-kx(-at+b)\longrightarrow|-k|^2\frac{E_x}{|-a|}$$

Power signals:

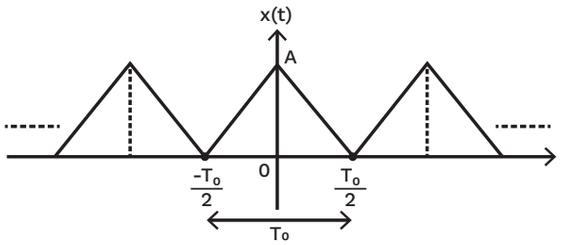
X(t) → Power of signal
→ Power of waveform
→ Mean square value of waveform

Power of Periodic Signals

$$P_x=\overline{x^2(t)}=MSV[x(t)]$$

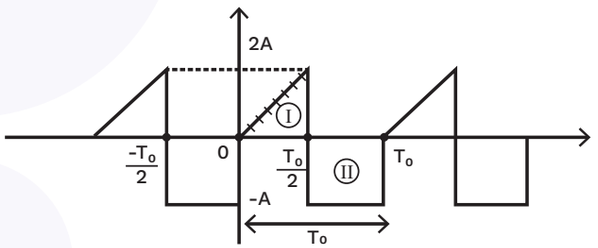
$$P_x=\overline{x^2(t)}=\frac{1}{T_0}\int_{\frac{-T_0}{2}}^{\frac{T_0}{2}}\left|x(t)\right|^2dt=\frac{E_{T_0}}{T_0}$$

Example:



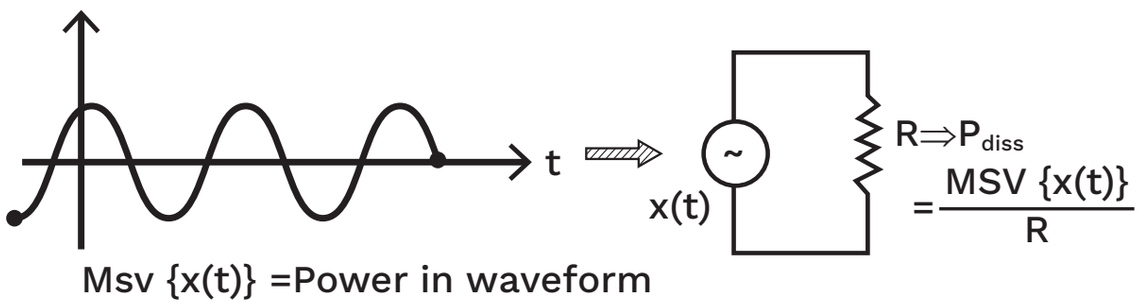
1. Periodic

2. $P_x=\overline{x^2(t)}=\frac{E_{T_0}}{T_0}=\frac{\frac{|A|^2}{3}\times T_0}{T_0}=\frac{|A|^2}{3}$



$$P_x=\overline{x^2(t)}=\frac{E_{T_0}}{T_0}=\frac{\frac{|2A|^2}{3}\times\frac{T_0}{2}+(-A)^2\times\frac{T_0}{2}}{T_0}$$

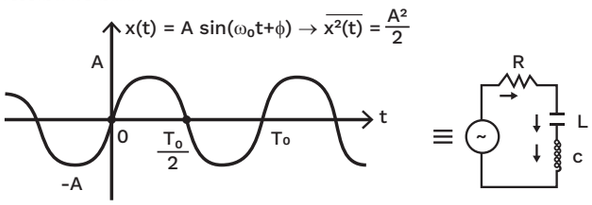
$P_x=?$



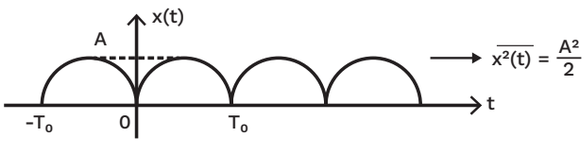
Msv {x(t)} =Power in waveform

Mean Square Value of Standard Waveforms

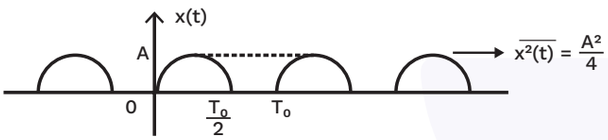
Sinusoidal:



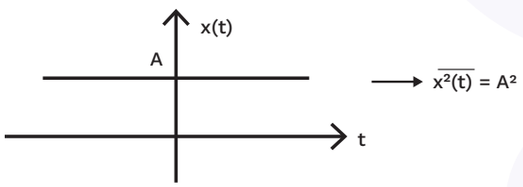
Full wave rectified version:



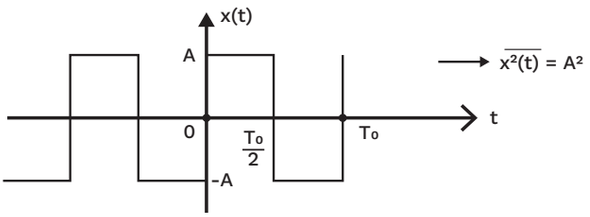
Half wave rectified version:



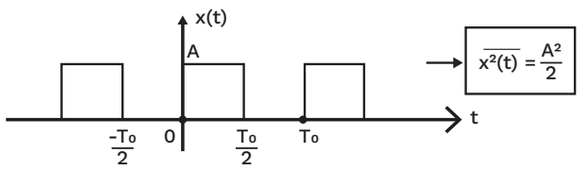
DC:

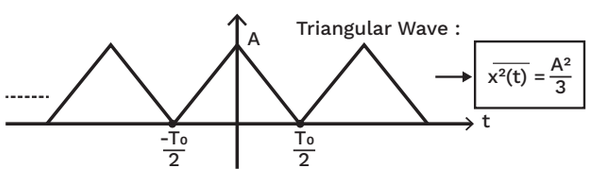


Square waveform:

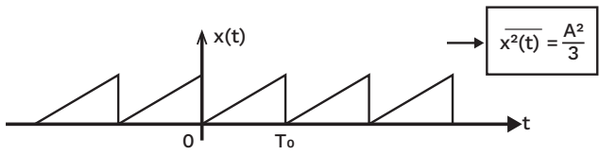


Rectangular pulse train:





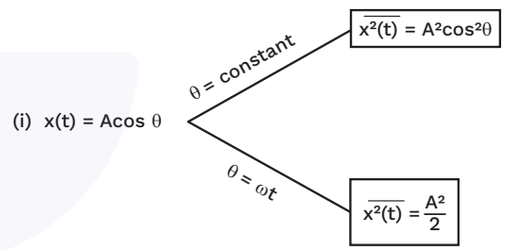
Sawtooth pulse train:



Properties:

$$\begin{cases} x(t) \to P_x & A\sin t \to \dfrac{A^2}{2} \\ x(-at+b) \to P_x & A\sin(-\omega_0 t+\phi) \to \dfrac{A^2}{2} \\ -kx(-at+b) \to |-k|^2 P_x & \end{cases}$$

Important observations:

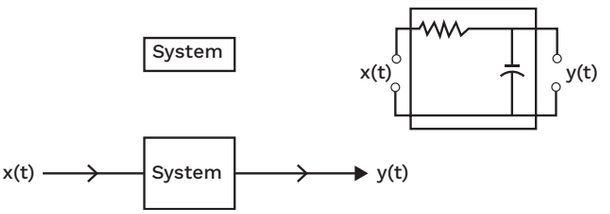


2. $x(t) = x_1(t) + x_2(t) + x_3(t) + ... + x_k(t)$

   $\{x_i\}_{i=1}^{k}$ → Set of orthogonal signals

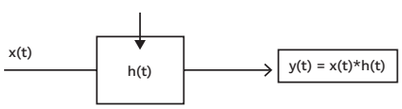
   $E_x = E_1 + E_2 + ... + E_k$

   $P_x = P_1 + P_2 + ... + P_k$

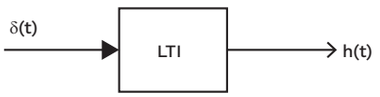


'Meaningful interconnection of components and devices which process input signal and generate the output signal'.

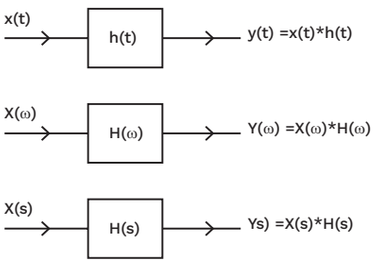
LTI Systems (Linear Time Invariant System)



$h(t)$: Unit impulse response of LTI system.



Representation of LTI system:



Standard results of convolution:

$x(t) * \delta(t) = x(t)$

$x(t-a) * \delta(t-b) = x(t-a-b)$

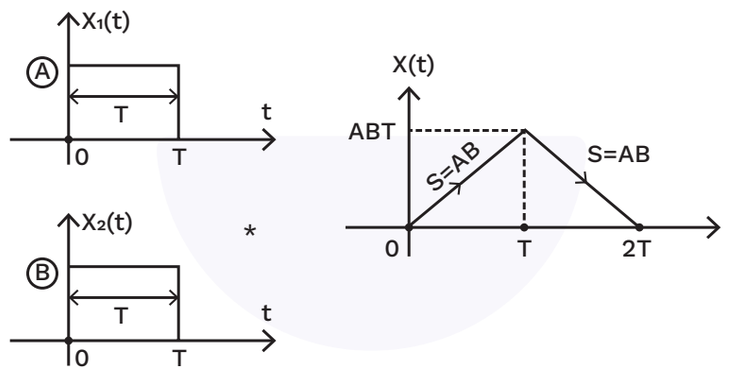
$u(t) * u(t) = r(t)$

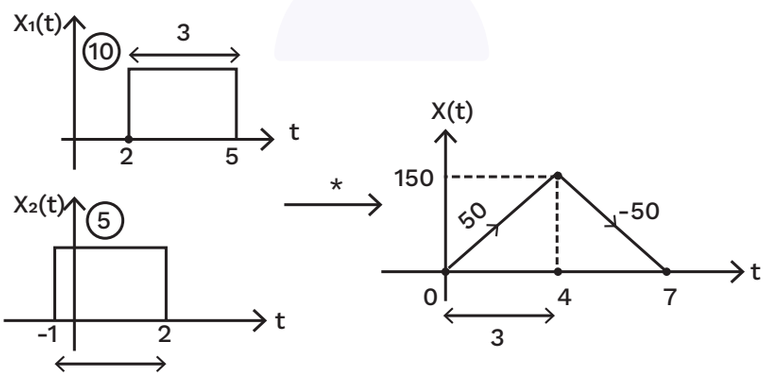
$u(t-a) * u(t-b) = r(t-a-b)$

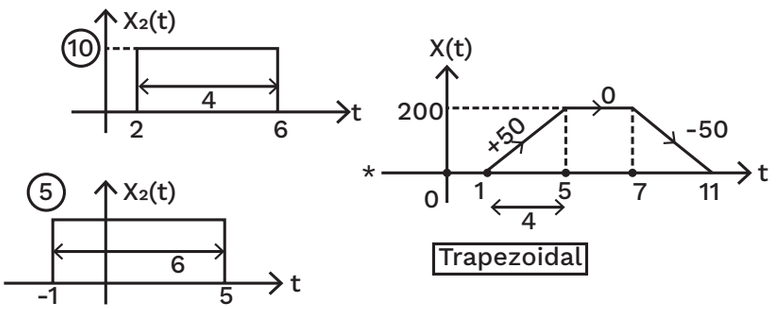
$u(t) * r(t) = p(t)$

$u(t-a) * r(t-b) = p(t-a-b)$

Convolution of two rectangular pulses of the same duration:



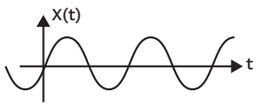




Trapezoidal

Time Domain Signal

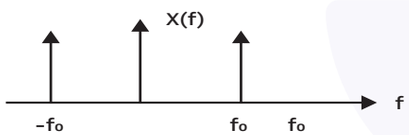
1. *'The description of a signal w.r.t time is known as time domain signal'.*
2. Horizontal axis (independent variable) = t
3. $x(t) = A\sin\omega_0 t$



4. Oscilloscope is a time domain device.

Frequency Domain signal

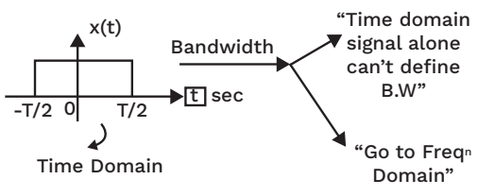
1. The description of a signal w.r.t. frequency is known as FD signals.
2. Horizontal axis (independent variable) is ω or f.
3. $X(f) = \delta(f - f_0) + \delta(f + f_0)$



4. Spectrum analyser is an FD device.

Important points

1. A time domain signal contains frequency. Information:
   a) Max. freq. component
   b) Bandwidth
   c) Frequency components
   d) Frequency range
2. By looking at the time domain graph, none of its frequency-related parameters cannot be calculated.
3. To calculate frequency-related parameters of the time domain signal, the time domain signal has to be converted into its equivalent frequency domain signal.



Tools to convert TDS into FDS

1. Continuous-time Fourier transform (Fourier transform)
2. Continuous-time Fourier series (Fourier series)

Fourier transform:

'It is a tool to convert continuous time (periodic + non-periodic) signal into frequentation domain'.

Relation:

$$x(t) \xleftrightarrow{F.T.} X(\omega)$$

$$x(t) \xleftrightarrow{F.T.} X(f)$$

Operator

$$F.T.\left[x(t)\right] = X(\omega)$$

$$x(t) \xleftrightarrow{F.T.} X(\omega) = \left|X(\omega)\right| e^{\angle X(\omega)}$$

1. Purpose: $X(\omega)$ *vs* $\omega \rightarrow$ *GRAPH*

2. Purpose: $\left|X(\omega)\right|$ *vs* $\omega \rightarrow$ *GRAPH*

Communication: x(t):Signals $\Rightarrow$ Real

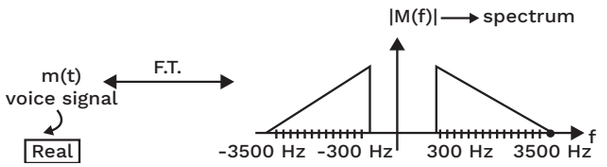
$$x(t) \leftrightarrow X(\omega) = \left|X(\omega)\right| e^{1\angle X(\omega)}$$

Real

Magnitude response: $\rightarrow$ Even

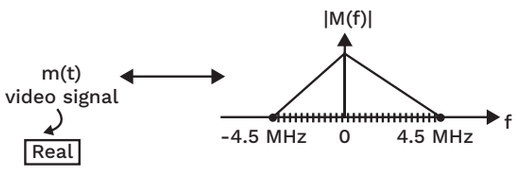
Phase response: $\rightarrow$ Odd

Voice signal: (300 Hz – 3500 Hz)



Bandwidth of $m(t) \Rightarrow (3500 - 300) = 3200\,Hz$

Video signal (0–4.5 MHz)



Bandwidth of m(t) $\Rightarrow$ (4.5 MHz - 0) = 4.5 MHz

Properties of FT

1. $x(t) \leftrightarrow X(\omega)$

$x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(\omega)$

$x(t+t_0) \leftrightarrow e^{+j\omega t_0} X(\omega)$

$x(t) \leftrightarrow X(f)$

$x(t-t_0) \leftrightarrow e^{-j2\pi f t_0} X(f)$

$x(t+t_0) \leftrightarrow e^{j2\pi f t_0} X(f)$

2. $x(t) \leftrightarrow X(\omega) \qquad x(t) \leftrightarrow X(f)$

$x(-t) \leftrightarrow X(-\omega) \quad x(-t) \leftrightarrow X(-f)$

3. $x(t) \leftrightarrow X(\omega) \qquad x(t) \leftrightarrow X(f)$

$e^{+j\omega_0 t} x(t) \leftrightarrow x(\omega - \omega_0)$

$e^{+j2\pi f_0 t} x(t) \leftrightarrow x(f - f_0)$

$e^{-j\omega_0 t} x(t) \leftrightarrow x(\omega + \omega_0)$

$e^{-j2\pi f_0 t} x(t) \leftrightarrow x(f + f_0)$

4. $x(t) \leftrightarrow X(\omega)$

$x(t)\cos\omega_0 t \leftrightarrow \dfrac{x(\omega-\omega_0)+x(\omega+\omega_0)}{2}$

$x(t) \leftrightarrow X(f)$

$x(t)\cos 2\pi f_0 t \leftrightarrow \dfrac{x(f-f_0)+x(f+f_0)}{2}$

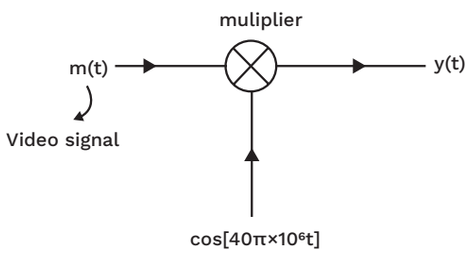
5. $x(t) \leftrightarrow X(\omega)$

$x(t)\sin\omega_0 t \leftrightarrow \left\{\dfrac{x(\omega-\omega_0)-x(\omega+\omega_0)}{2j}\right\}$

$x(t) \leftrightarrow X(f)$

$x(t)\sin 2\pi_0 t \leftrightarrow \left\{\dfrac{x(f-f_0)-x(f+f_0)}{2j}\right\}$

Example:



Plot spectrum of y(t)

Solution:

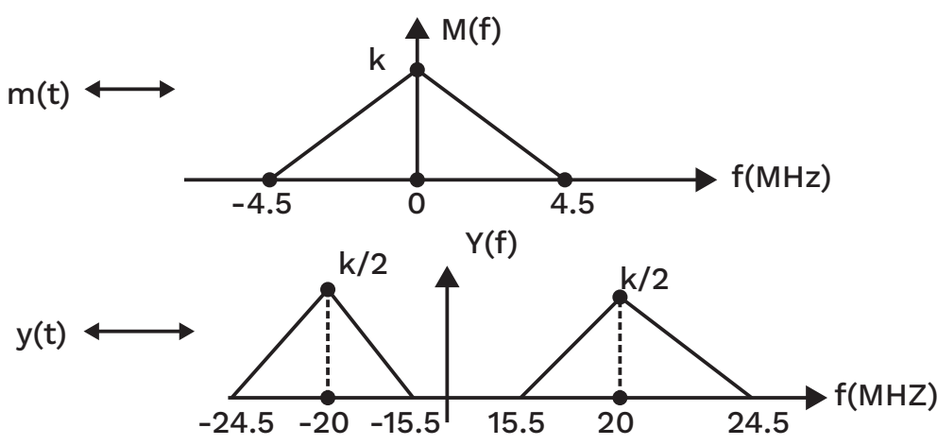
$y(t) = m(t)\cos(40\pi \times 10^6 t)$

$f_0 = 20\,MHz$

$m(t) \leftrightarrow M(f)$

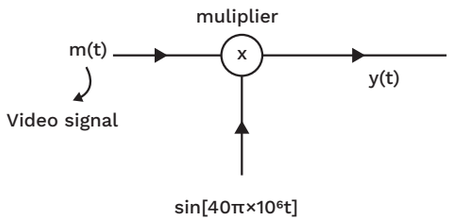
$m(t)\cos(2\pi(20\times 10^6 t)) \leftrightarrow$

$\dfrac{M(f-20\,MHz)+M(f+20MHz)}{2}$



Bandwidth of $y(t) \Rightarrow (24.5 - 15.5) = 9\,MHz$

Example:



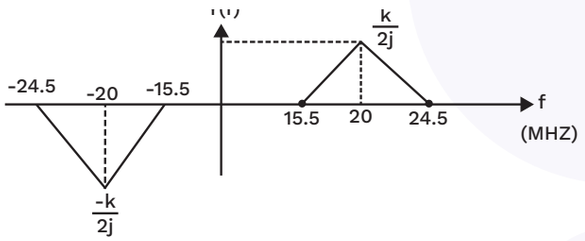
'Plot spectrum of y(t)'.

Solution:

$$m(t) \leftrightarrow M(f)$$

$$y(t) = m(t)\sin\left(2\pi\left(20\times10^{6}t\right)\right) \leftrightarrow$$

$$\frac{M(f-20\,MHz)}{2j} - \frac{M(f+20\,MHz)}{2j} = Y(f)$$



Bandwidth of y(t) → 9 MHz

Conversion of $X(\omega)$ vs ω to X(f) vs f:

1. $X(\omega) = \omega^2$

   Dummy variable

   $X(f) = (2\pi f)^2 = 4\pi^2 f^2$

2. $X(\omega) = \sin\omega$

   $X(f) = \sin(2\pi f)$

3. $X(\omega) = \delta(\omega)$

   $X(f) = \delta(2\pi f)$

   $X(f) = \frac{1}{2\pi}\delta(f)$

a) *'Horizontal axis always divided by 2π'.*

b) *'Vertical axis containing impulses divided by 2π'.*

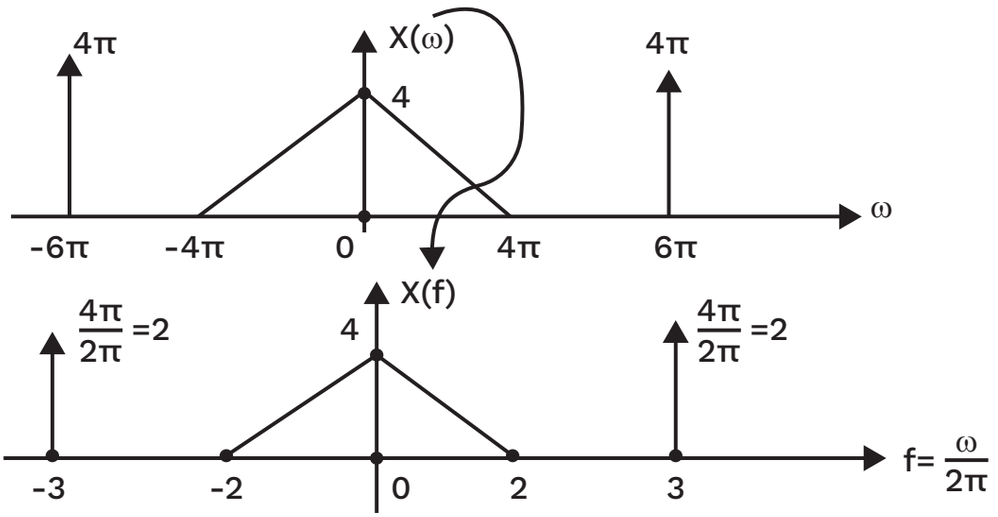
c) Zero frequency value remains the same.

d) During conversion, shape does not change.

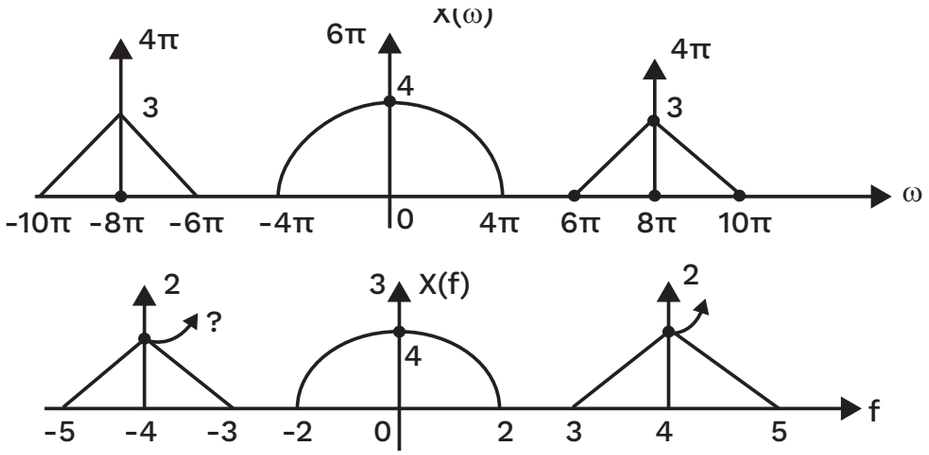
$X(\omega) = \omega^2 \xrightarrow{\omega=0}$

$X(f) = 4\pi^2 f^2 \xrightarrow{f=0}$

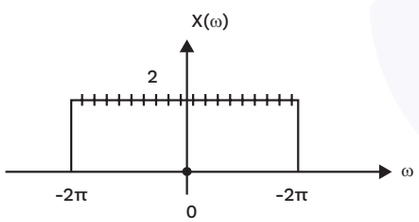
Example:

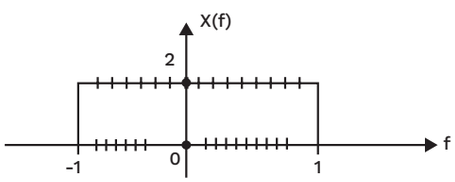


Example:



Example:





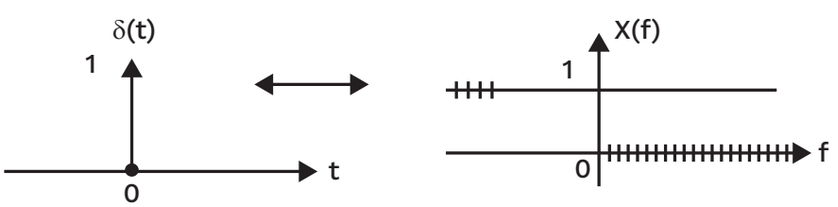
Fourier Transform of Standard Signals

1. $x(t) = \delta(t) \xleftrightarrow{F.T.} X(\omega) = 1$

   Or

   $X(f) = 1$

   Bandwidth of $\delta(t) \rightarrow (\infty - 0) = \infty Hz$



$f \varepsilon [0, \infty] \rightarrow 0 \leq f < \infty$

2. $x(t) = A\cos\omega_0 t \leftrightarrow X(\omega) =$

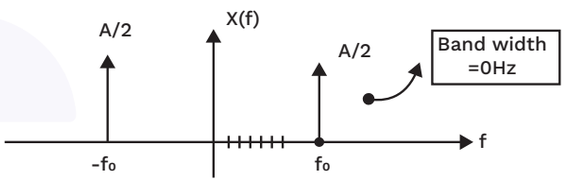
   $A\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$

   Or

   $X(f) = \frac{A}{2}[\delta(f - f_0) + \delta(f + f_0)]$

   Periodic

   $x(t) = A\cos\omega_0 t$



Single discrete frequency = f_0 *Hz*

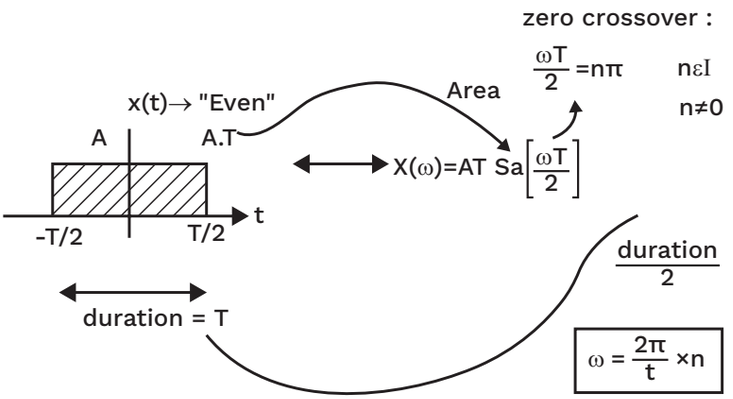
Bandwidth of x(t) = $f_0 - f_0 = 0\,Hz$

3. $x(t) = A\sin\omega_0 t \leftrightarrow x(\omega) =$

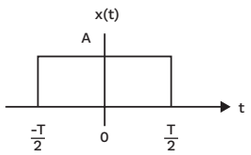
   $\frac{A\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$

$$X(f) = \frac{A}{2j}\left[\delta(f - f_0) - \delta(f + f_0)\right]$$

Rectangular pulse:



$$x(t) \rightarrow E_x = A^2T$$

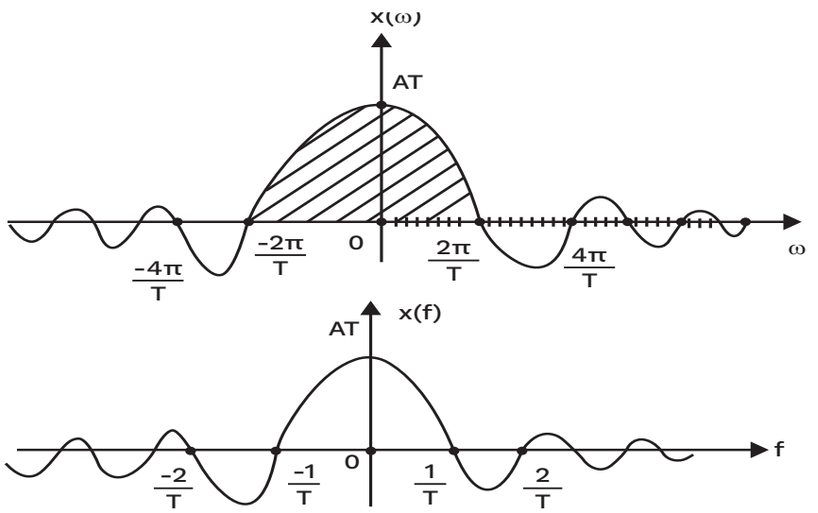


$0 \le f < \infty$

Except

$f = \frac{n}{T} : n \neq 0$

Bandwidth of $x(t) = (0 - 0)Hz$



Significant frequentation:

Main lobe freq.

$$\omega \in \left[0, \frac{2\pi}{T}\right]$$

$$f \in \left[0, \frac{1}{T}\right]$$

$$E_x = \int_{-\infty}^{\infty} x^2(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |x(\omega)|^2\,d\omega = \int_{-\infty}^{\infty} |x(f)|^2\,df$$

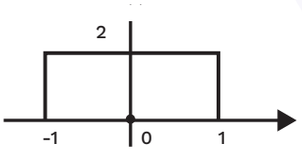
$$\int_{-1/T}^{1/T} |x(f)|^2\,df = 90\% \text{ to } 95\% \text{ of } \left[A^2T\right]$$

Example: $X(\omega) = \frac{4\sin\omega}{\omega}$

$$X(\omega) = 4\,Sa\left[1\omega\right]$$

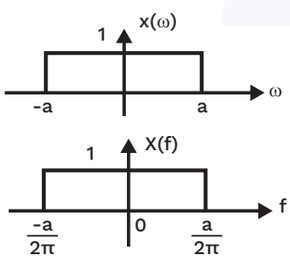
$$\frac{duration}{2} = 1$$

Area = 4



(4)

$x(t) = \frac{\sin at}{\pi t}$ ⟷

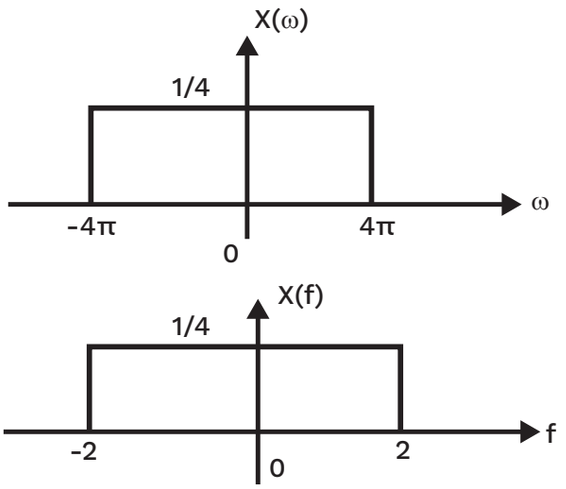


Bandwidth of x(t) $\Rightarrow$ (a – 0) rad/sec = $\frac{a}{2\pi} Hz$

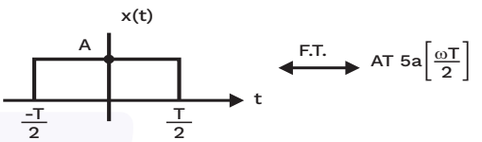
$$x(t) = \sin(4t)$$

$$= \frac{\sin(4\pi t)}{4\pi t}$$

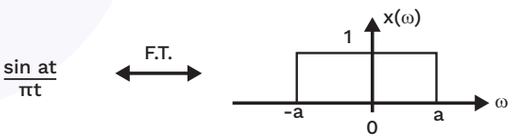
$$= \frac{1}{4}\left\{\frac{\sin(4\pi)t}{\pi t}\right\}$$



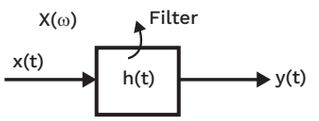
1.



2.



Convolution Property and (Filtration)



$$y(t) = x(t) * h(t) \Rightarrow Y(f) = X(f)H(f)$$

$$y(\omega) = X(\omega)H(\omega)$$

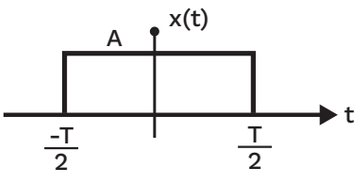
Filtered version of x(t)

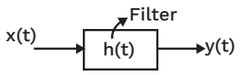
Example:



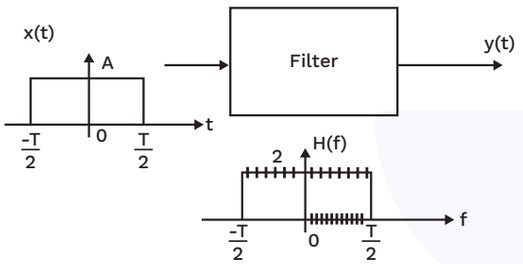
$$f_c = \frac{1}{T} Hz, PBG = 2$$

Filtered version of x(t) = ?



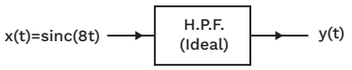


1. $y(t) = x(t) * h(t)$
2. $y(\omega) = x(\omega)H(\omega)$ or $y(f) = X(f)H(f)$



Note: *'Low pass filter is a device used to convert band unlimited signal into band limited signal'.*

Example: $x(t) = \sin(8t)$

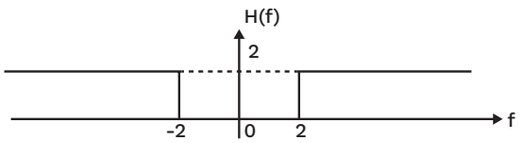


$f_c = 2Hz$

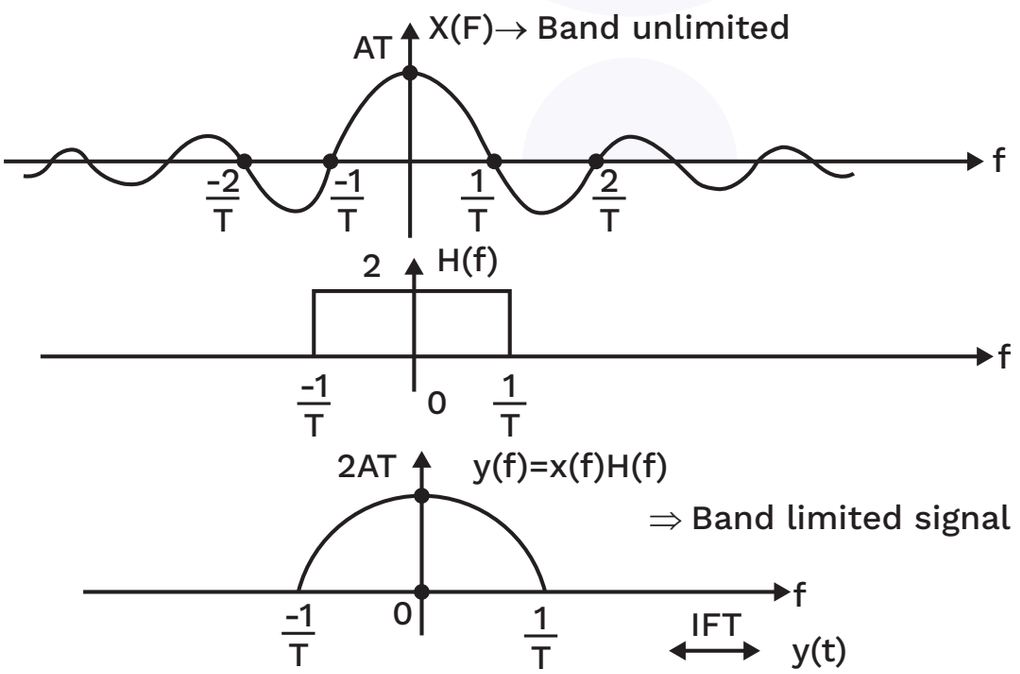
PBG = 2

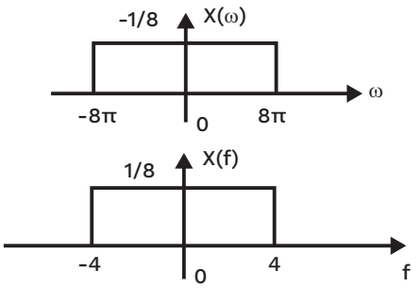
Plot the spectrum of y(t) and calculate the BW of y(t).

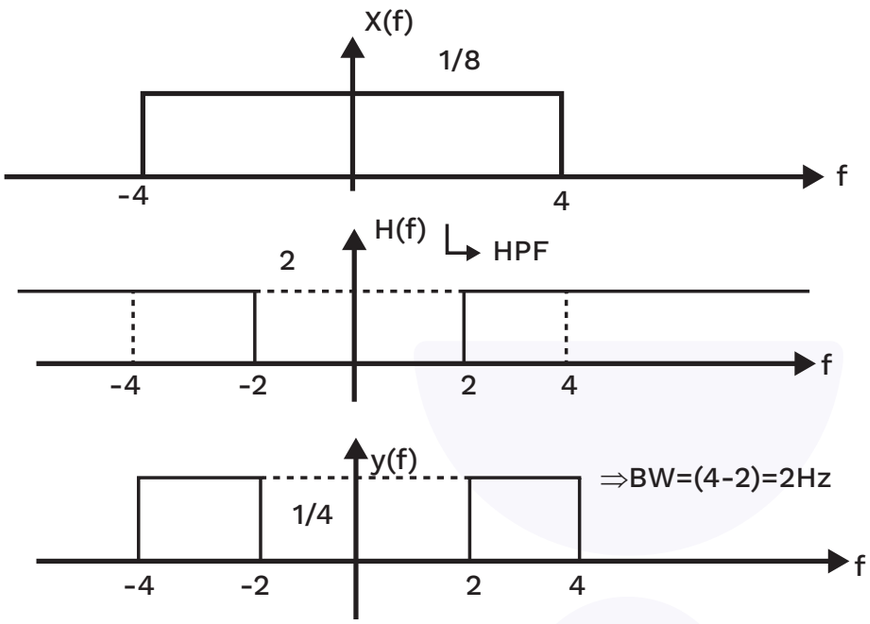
Solution:



$$\text{sinc}(8t) = \frac{\sin 8\pi t}{8\pi t} = \frac{1}{8}\left\{\frac{\sin 8\pi t}{\pi t}\right\}$$







Absolute bandwidth of x(t):

(Highest +ve frequency at which component is present) - (Lowest +ve frequency at which component is present).

Multiplication property:

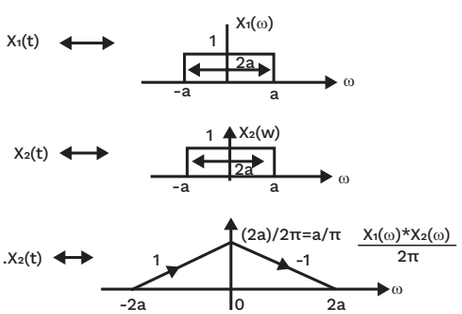
$$x_1(t).x_2(t) \leftrightarrow \frac{1}{2\pi}\left[x_1(\omega) * x_2(\omega)\right]$$

Or

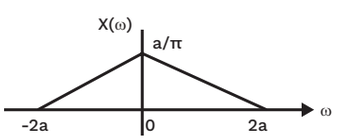
$$\left[x_1(f) * x_2(f)\right]$$

$$x(t) = \left(\frac{\sin at}{\pi t}\right)^2 \leftrightarrow X(\omega) = ?$$

$$\left(\frac{\sin at}{\pi t}\right)_{x_1(t)} \times \left(\frac{\sin at}{\pi t}\right)_{x_2(t)} \leftrightarrow \frac{1}{2\pi} x_1(\omega) * x_2(\omega)$$



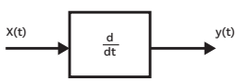
$$\left(\frac{\sin at}{\pi t}\right)^2 \xleftrightarrow{F.T.}$$



Differentiation property:

$$x(t) \leftrightarrow X(\omega)$$

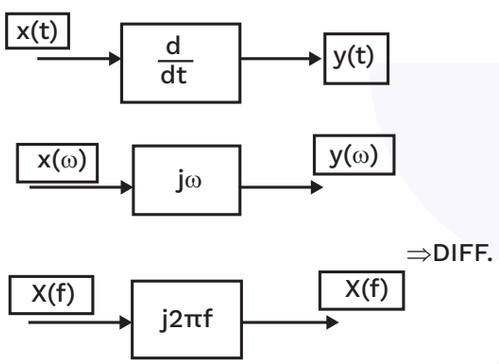
$$\frac{dx(t)}{dt} \leftrightarrow j\omega X(\omega)$$



$$y(t) = \frac{dx(t)}{dt}$$

$$y(\omega) = j\omega X(\omega)$$

$$H(\omega) = j\omega \quad or \quad H(f) = j2\pi f$$



Fourier Transform of Gaussian

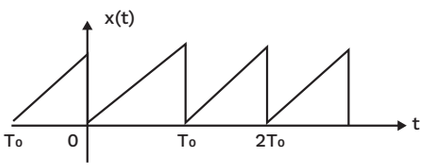
$$X(t) = e^{-at} \leftrightarrow X(\omega) = \sqrt{\frac{\pi}{a}} e^{\frac{-\omega^2}{4a}}$$

$$X(t)e^{-\pi t^2} \leftrightarrow X(\omega) = X(\omega) = e^{\frac{-\omega^2}{4\pi}}$$

$$x(t)e^{-\pi t^2} \leftrightarrow X(f) = e^{-\pi f^2}$$

Fourier series:

1. Fourier series is used to determine frequency components of time domain, non-sinusoidal periodic signal.



$$b_0 = 0$$

$$x(t) = a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + \ldots$$

$$+b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + \ldots$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos n\omega_0 t + b_n \sin n \cos t\right)$$

'It is a method to write non-sinusoidal periodic signal as the linear combination of mutually orthogonal, harmonically related sinusoids'.

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right)$$

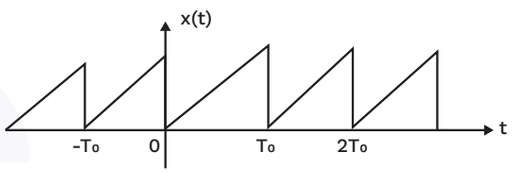
$$x(t) = \sum_{n=0}^{\infty} r_n \cos\left(n\omega_0 t - \phi_n\right)$$

$$a_n = r_n \cos \phi_n$$

$$b_n = r_n \sin \phi_n$$

$$r_n = \sqrt{a_n^2 + b_n^2}$$

$$\phi_n = \tan^{-1} b_n / a_n$$



$$x(t) = r_0 + r_1 \cos(\omega_0 t - \phi_1) + r_2 \cos(2\omega_0 t - \phi_2) + \ldots$$

Frequency components absent in x(t) ⇒ Will be those harmonics for which $r_n = 0$

Note: To calculate frequency components of non-sinusoidal periodic signal:

S.1 → Calculate $r_n = \sqrt{a_n^2 + b_n^2}$

$a_n = 2/T_0 \int_{T_0} x(t) \cos n\omega_0 t \, dt$

$b_n = 2/T_0 \int_{T_0} x(t) \sin n\omega_0 t \, dt$

S.2 → $r_n = 0 \Rightarrow$ *Calculate* n

S.3 → Frequencies absent will be $\Rightarrow nf_0$

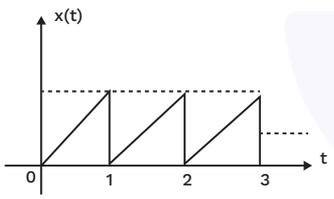
Example: $r_n = \text{sinc}\left(\frac{n}{3}\right)$

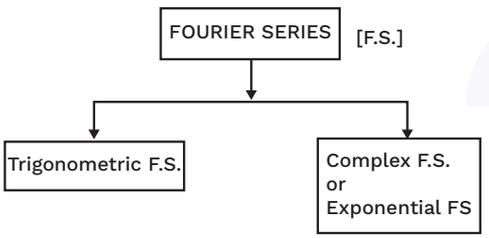
$r_n = 0 \rightarrow \frac{n}{3} = k \qquad k \in I$

$n = 3k \quad k \neq 0$

n = 3, 6, 9 -----

Freqⁿ absent





FREQ Components Absent in Non-Sinusoidal Periodic Signal

Harmonics for which

$\rightarrow r_n = 0$ or $|C_n| = 0$

or $C_n = 0$

$C_n = |C_n| e^{j\angle C_n}$

$\downarrow 0 \quad \downarrow 0$

$$C_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega_0 t} dt \rightarrow C_o = \frac{1}{T_0} \int_{T_0} x(t) dt = \frac{\text{Area}_{T_0}}{T_0}$$

Half Wave Symmetry

$$x(t) = -x\left(t + \frac{T_0}{2}\right)$$

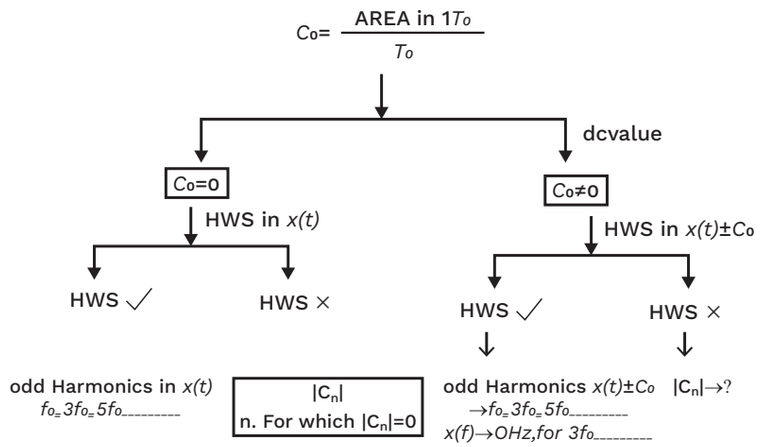
Mathematical exp: $x\left(t + \frac{T_0}{2}\right) = -x(t)$

Graphical understanding:

$$t = 0^+ \Rightarrow x\left(\frac{T_0^+}{2}\right) = -x(0^+)$$

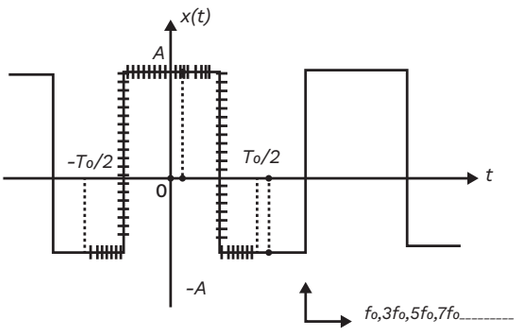
$$x(0^+) = -x\left(\frac{T_0^+}{2}\right)$$

Procedure to check hws:



Some standard signals:

SYMM square wave:



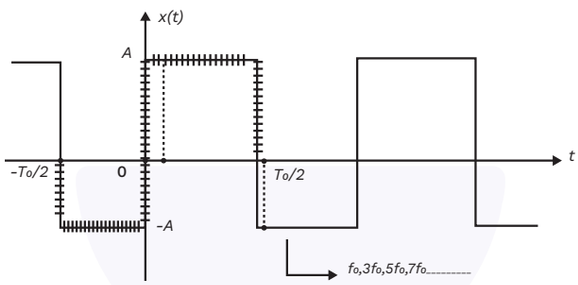
$$\Rightarrow C_0 = \frac{A_{T_0}}{T_0} = 0$$

$$\Rightarrow x(0^+) = -x\left(\frac{T_0^+}{2}\right)$$

$$\downarrow \quad \downarrow$$

$$A \quad HWS$$

Asymmetrical-square waveform:



$$\rightarrow C_o = 0$$

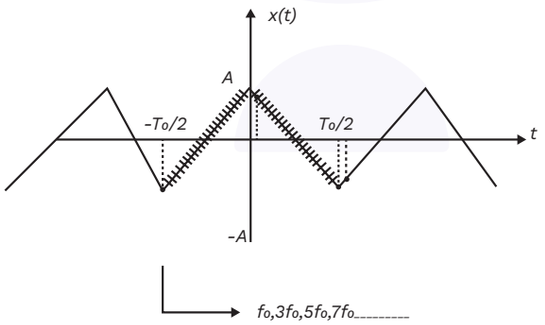
$$\rightarrow x\left(0^+\right) = +A$$

$$x\left(\frac{T_0^+}{2}\right) = -A$$

$$\downarrow$$

$$HWS$$

SYMN triangular wave:



$$\rightarrow C_0 = 0$$

$$x(0^+) = A - \in$$

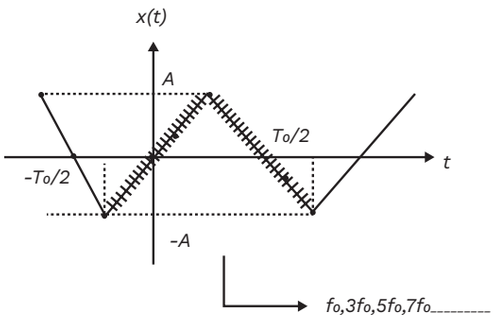
$$x\left(\frac{T_0^+}{2}\right) = -A + \in$$

$$x\left(0^+\right) = -x\left(\frac{T_0^+}{2}\right)$$

$$\downarrow$$

$$HWS$$

Asymm triangular wave:



$$C_0 = 0$$

$$x\left(0^+\right) = \in$$

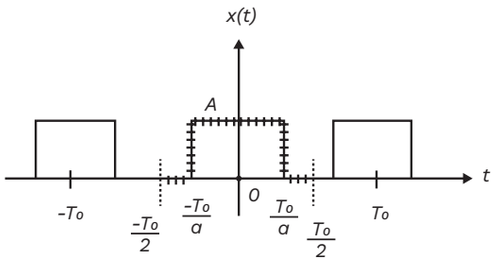
$$x\left(\frac{T_0^+}{2}\right) = -\in$$

$$\boxed{x\left(0^+\right) = -x\left(\frac{T_0^+}{2}\right)}$$

$$\downarrow$$

$$HWS$$

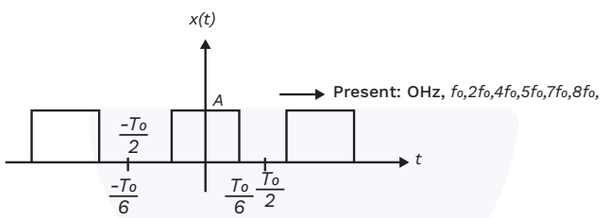
Rectangular pulse train:



$C_0 \neq 0$

Frequencies absent will be: $K = +ve$ Integer

Harmonics for which $n = \frac{Ka}{2}$



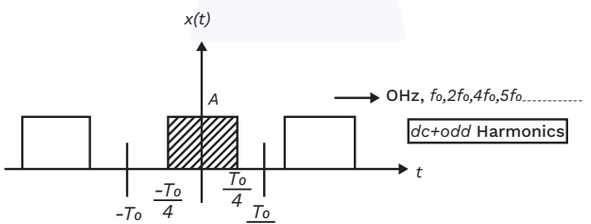
$\nearrow 3f_0, 6f_0, 9f_0 \ldots$

Frequncies Absent:

$a = 6$

$n = \frac{Ka}{2} = \frac{K}{2} \times 6 = 3K$

$3, 6, 9 \ldots$

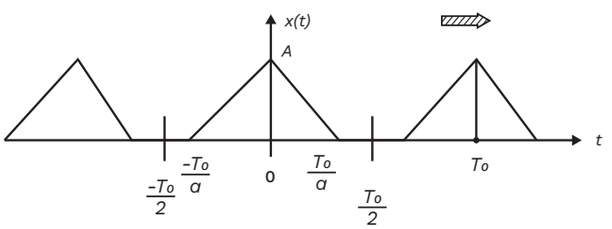


Frequencies absent: $n = \frac{Ka}{2}$

$a = 4 \quad \Rightarrow \quad n = 2k$

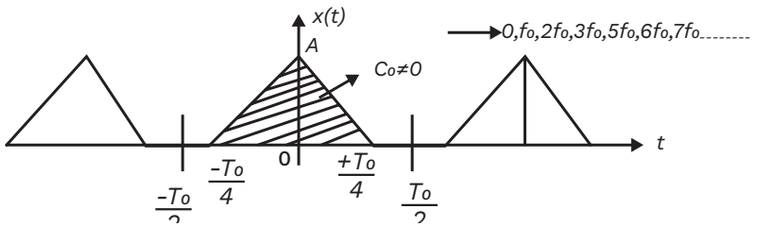
$2f_0, 4f_0, bf_0 \ldots$

Triangular pulse train:



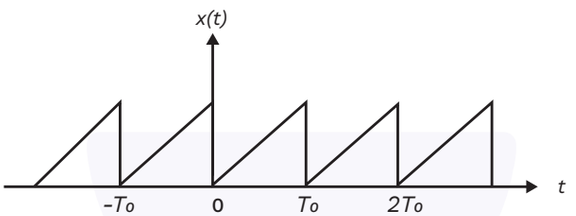
'Frequencies absent will be the harmonics'.

$\boxed{n = Ka}$ $\boxed{K = +ve \text{ integer}}$

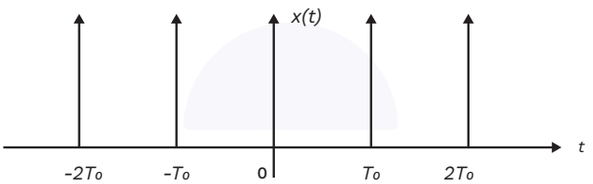


Frequency absent: $\Rightarrow N = Ka = 4K$

Periodic signal containing all discrete frequencies:



$\Rightarrow 0, f_0, 2f_0, 3f_0, 4f_0 \ldots$



$\Rightarrow 0, f_0, 2f_0, 3f_0, 4f_0 \ldots$

Example 1: A sawtooth pulse train of the time period 100μsec, has max. frequency component up to 5thharmonic. Calculate *BW* of sawtooth pulse train.

$$\boxed{f_0 = \frac{1}{100\mu \sec} = 10KHz}$$

$x(+) \rightarrow 0, f_0, 2f_0, 3f_0, 4f_0, 5f_0$

$BW = (5f_0) - 0 = 5f_0$hz

50KHz.

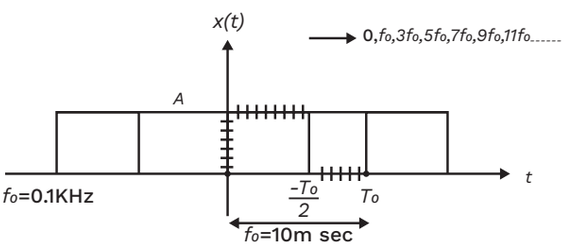
Example 2: An Asymm. square wave has a time period of 100 μsec. Maximum frequency is up to 5th harmonic. Calculate the bandwidth.

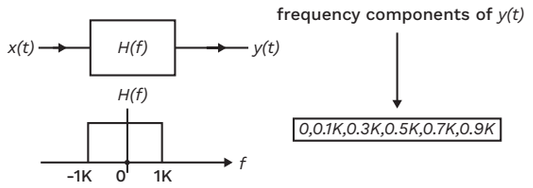
Solution:

$x(t) \rightarrow f_0, 3f_0, 5f_0 \ldots$

$BW = (5f_0) - f_0 = 4f_0 = 40$KHz

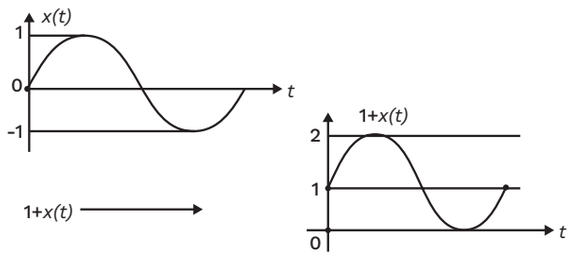
Example 3:



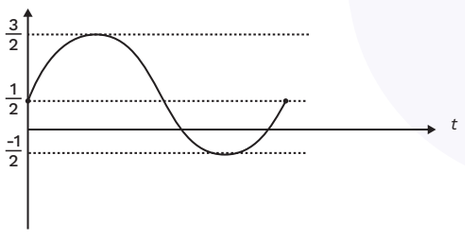


Concept of envelop:

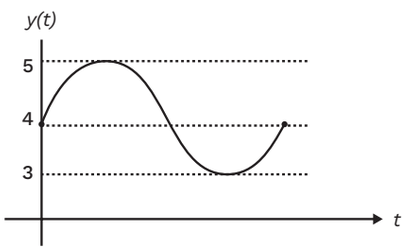
Operations on the dependent variable:



$$y(t) = \frac{1}{2} + x(t)$$



$$y(t) = 4 + x(t)$$



Envelop of a signal:

'The shape obtained by tracing positive peaks of a signal'.

Rep: $E(t) vs\ t$.

Procedure to plot graph of signal:

$x(t) = A(t)\cos w_0 t$ & $x(t) = A(t)\sin w_0 t$

S.1 Plot A(t) in reference form.

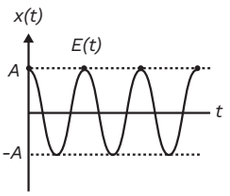
S.2 Take its mirror image w.r.t horizontal axis.

S.3 $X(t)\big|_{t=0} \rightarrow$ Starting point

S.4 Draw a sinusoidal shape.

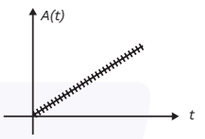
Example 1:

1. A(t) = A
2. ✓
3. $x(t)\big|_{t=0} = A$



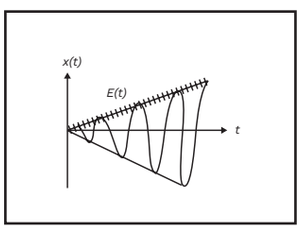
$\boxed{A(t) > 0} \longrightarrow \boxed{E(t) = A(t)}$

Example 2: $x(t) = t\cos t \xrightarrow{t=0} 0$



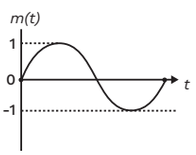
A(t) = t $\Rightarrow$

$A(t) > 0 \longrightarrow \boxed{E(t) = A(t)}$

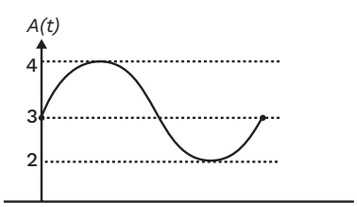


Example 3:

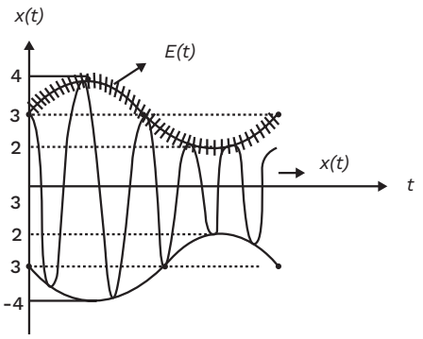
$x(t) = \underline{(3 + m(t))}\cos\omega_c t \xrightarrow{t=0} 3 + m(0) = 3$



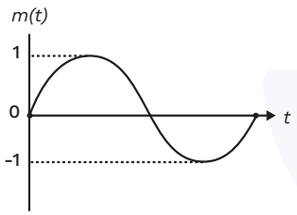
A(t) = 3 + m(t)



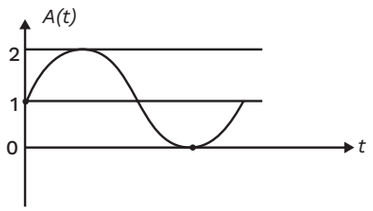
$A(t) > 0 \Rightarrow E(t) = A(t)$



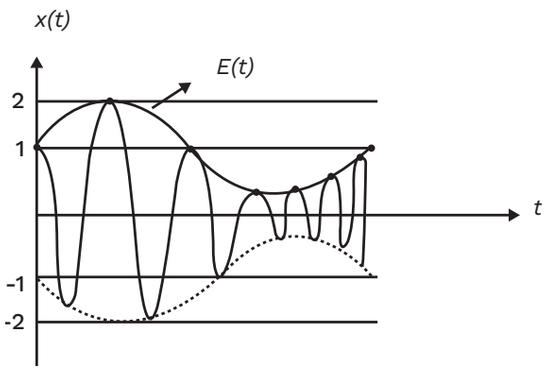
Example 4: $X(t) = (1 + m(t))\cos\ \omega_c t \xrightarrow[t=0]{} 1$



$A(t) = 1 + m(t)$

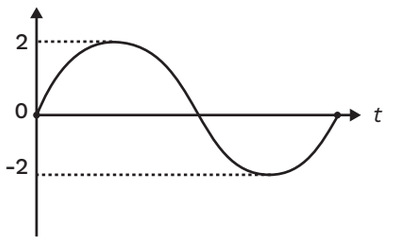


$A(t) \geq 0 \Rightarrow \boxed{E(t) = A(t)}$

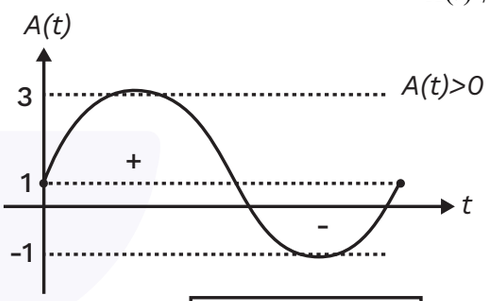


Example 5:

$x(t) = (1 + m(t))\cos\ \omega_c t \xrightarrow[t=0]{} 1$

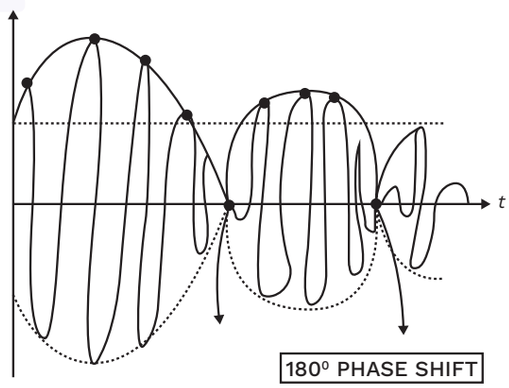


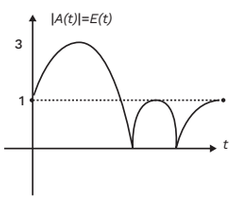
$A(t) \not> 0$



$\boxed{A(t)>0 \rightarrow E(t) \neq A(t)}$

$\boxed{A(t) \not> 0 \longrightarrow E(t) \neq A(t)}$





Conclusion

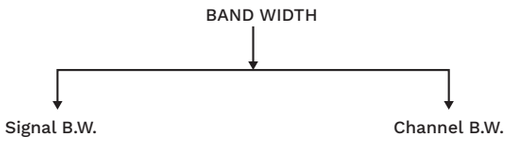
1. $x(t) = A(t)\cos\omega_c t$ or $A(t)\sin\omega_c t$

Envelop of $x(t) \Rightarrow E(t) = A(t) : A(t) \geq 0$

$E(t) = |A(t)| : A(t) \not> 0$

2. $x(t) = A(t)\cos\omega_c t$

Envelop of $x(t) \Rightarrow E(t) = |A(t)|$



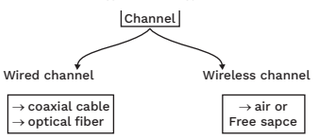
Signal bandwidth:

'Range of positive frequencies occupied by spectrum of the signal'.

Channel bandwidth:

'Range of frequencies allowed by the channel'.

Channel: *'Medium through which message propagates from* T_x to R_x ".

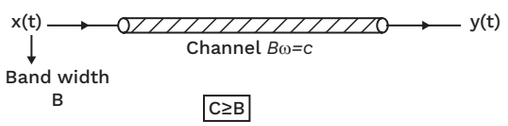


→ coaxial cable
→ optical fiber

↓

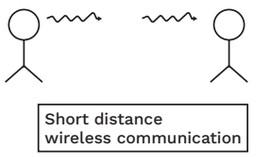
$Bw = \text{finite}$

Relation between C and B:



Types of Communication

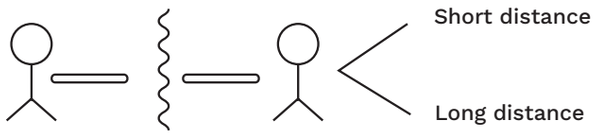
1. Wireless physical communication

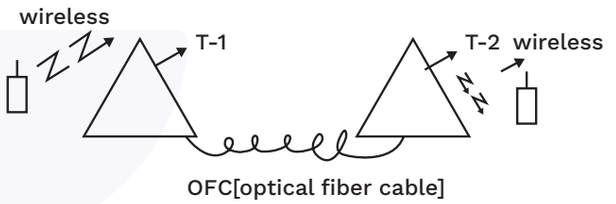


2. Wired communication



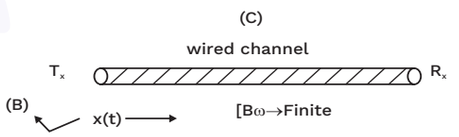
3. Wireless communication





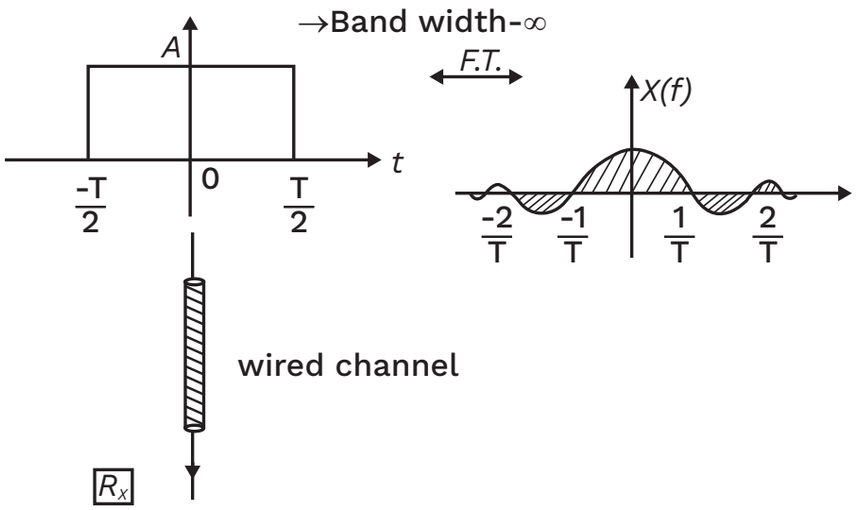
Electronic communication:

1. T_x and R_x are connected by a wired channel.



2. All wired channels have finite BW.
3. For faithful T_x and R_x signal through the wined channel (either long or short). $C \geq B$

Example: What happens if $x(t)$ having $B \to \infty$ is transmitted through the wired channel?



- Signals having bandwidth cannot be Txed completely through a wired channel of finite bandwidth.
- Receivers of the communication system can recover the message signal even if significant frequencies of the signal are Txed to them.
- In the communication system, there should always be transmission of significant frequencies.
- **Band limiting:** Band limiting is the process by which band unlimited signal is converted into band limited signal.
- *'Lowpass filter is used for band limiting'.*

Baseband signal:

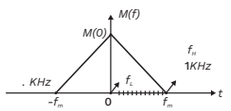
1. *'Baseband signals are naturally existing message signals'.*
2. Baseband signals are practical message signals.
3. Baseband signals have a lower cut-off frequency.

Either OHZ or very closed to OHZ

Bandpass signals: Fabricated signals

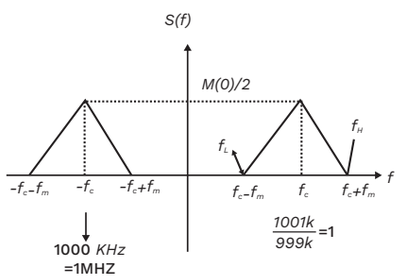
Bandpass signals are obtained from the baseband signal by shifting the baseband signal to very high frequencies.

One of the possible methods to convert a baseband signal into band pass signals is multiplying the baseband with a high-frequency carrier.



BASEBAND

$m(t)$ ⟷



Low pass signal Baseband

Baseband S/G	Frequency	Significant FEQ
Voice	300Hz ⟶ ∞	300Hz ⟶ 3.5KHz
Audio	20Hz ⟶ ∞	20*Hz* ⟶ 20*kHz*
Video	0Hz ⟶ ∞	0 ⟶ 4.54*Hz*

↗ Bandpass

$S(t) = m(t)\cos\omega_c t$ $\longleftrightarrow$

$f_c >>> f_m$

$f_c - f_m >>>> 0 \rightarrow$

$-f_c + f_m <<<< 0$

Signal	Signal BW	Wired	Wireless
Baseband	Finite	✓	?
Bandpass	Finite	✓	?

Wideband signals: f_L, f_H $\boxed{\frac{f_H}{f_L} >>> 1}$

Voice
Audio $\Rightarrow$ $\frac{f_H}{f_L} >> 1$ $\longrightarrow$ BASEBAND Sig
Video

Narrowband signal $\boxed{\frac{f_H}{f_L} \approx\sim 1}$